

Matrices

1- Square Matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

2- Rectangular Matrix

$$\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

Matrices

3- Column Vector

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

4- Row Vector

$$[a \quad b \quad c]$$

Matrices

5- $m \times n$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & \dots & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & \dots & \dots & \dots & a_{2n} \\ \vdots & & \vdots & & \vdots & & \vdots \\ \vdots & & \vdots & & \vdots & & \vdots \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & \dots & \dots & \dots & a_{mn} \end{bmatrix}$$

Matrices

Special Matrices

1- Zero Matrix

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2- Identity Matrix

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrices

Special Matrices

3- Diagonal Matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

4- Tridiagonal Matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 7 & 3 & 5 \\ 0 & 8 & 4 \end{bmatrix}$$

Matrices

Special Matrices

5- Upper Triangular Matrix

$$A = \begin{bmatrix} 2 & 1 & 5 \\ 0 & 3 & 6 \\ 0 & 0 & 4 \end{bmatrix}$$

6- Lower Triangular Matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 6 & 5 & 4 \end{bmatrix}$$

Transpose of Matrices

$$A = \begin{bmatrix} 2 & 1 & 5 \\ 7 & 3 & 6 \\ 8 & 9 & 4 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 7 & 8 \\ 1 & 3 & 9 \\ 5 & 6 & 4 \end{bmatrix}$$

Transpose of Matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$(A + B)^T = A^T + B^T$$

$$(A B)^T = B^T A^T$$

Transpose of Matrices

Symmetric Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 5 \end{bmatrix}$$

$$A^T = A$$

Skew – Symmetric Matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$$

$$A^T = -A$$

Addition and Multiplication

Addition

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} =$$

Addition and Multiplication

Addition

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a + e & b + f \\ c + g & d + h \end{bmatrix}$$

Addition and Multiplication

Addition

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a + e & b + f \\ c + g & d + h \end{bmatrix}$$

Multiplication by scalar k

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

Multiply two matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} =$$

Multiply two matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Multiply two matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

$$\begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} =$$

Multiply two matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

$$\begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ea + fc & eb + fd \\ ga + hc & gb + hd \end{bmatrix}$$

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix} =$$

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix}_{3 \times 4} =$$

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix}_{3 \times 4} =$$

2 × 3

3 × 4

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix}_{3 \times 4} =$$

condition

2×3

3×4

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix}_{3 \times 4} = \begin{bmatrix} \emptyset & \emptyset & \emptyset & \emptyset \\ \emptyset & \emptyset & \emptyset & \emptyset \end{bmatrix}_{2 \times 4}$$

condition
2 x 3 3 x 4
result

Multiply two matrices

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 1 & 2 & 3 \end{bmatrix}_{3 \times 4} = \begin{bmatrix} \emptyset & \emptyset & \emptyset & \emptyset \\ \emptyset & \emptyset & \emptyset & \emptyset \end{bmatrix}_{2 \times 4}$$

condition
 2×3 3×4
result

$$\begin{bmatrix} (1 * 1 + 2 * 5 + 3 * 9) & (1 * 2 + 2 * 6 + 3 * 1) & (1 * 3 + 2 * 7 + 3 * 2) & (1 * 4 + 2 * 8 + 3 * 3) \\ (4 * 1 + 5 * 5 + 6 * 9) & (4 * 2 + 5 * 6 + 6 * 1) & (4 * 3 + 5 * 7 + 6 * 2) & (4 * 4 + 5 * 8 + 6 * 3) \end{bmatrix}$$

The Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longrightarrow \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = (1 \cdot 4) - (2 \cdot 3) = 4 - 6 = -2$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \quad \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

$$= 1 \begin{vmatrix} 5 & 6 \\ 8 & 1 \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

$$= 1 \begin{vmatrix} 5 & 6 \\ 8 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 1 \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

$$= 1 \begin{vmatrix} 5 & 6 \\ 8 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

The Determinant

$$= 1 \begin{vmatrix} 5 & 6 \\ 8 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= 1 (5 \cdot 1 - 6 \cdot 8) - 2 (4 \cdot 1 - 6 \cdot 7) + 3 (4 \cdot 8 - 5 \cdot 7)$$

The Determinant

$$= 1 \begin{vmatrix} 5 & 6 \\ 8 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= 1 (5 \cdot 1 - 6 \cdot 8) - 2 (4 \cdot 1 - 6 \cdot 7) + 3 (4 \cdot 8 - 5 \cdot 7)$$

$$= 1 (5 - 48) - 2 (4 - 42) + 3 (32 - 35)$$

$$= 1 (-43) - 2 (-38) + 3 (-3) = -43 + 76 - 9 = 24$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{vmatrix}$$

The Determinant

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{bmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 1 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= (1 \cdot 5 \cdot 1) + (2 \cdot 6 \cdot 7) + (3 \cdot 4 \cdot 8) - (3 \cdot 5 \cdot 7) - (1 \cdot 6 \cdot 8) - (2 \cdot 4 \cdot 1)$$

$$= 5 + 84 + 96 - 105 - 48 - 8 = 24$$